

The published datasets are processed and anonymised versions of the original healthcare survey data. Direct identifiers and sensitive participant information have been removed or transformed prior to release.

Pre-processing Data

In the results table, the data is first pre-processed into normalised and unnormalised data. After the data is cleaned, the data goes through the min-max process to normalise it. Both the normalised and unnormalised data goes through encoding so that the data is uniform.

The raw pre-processing data is not included in this repository because the original healthcare survey data contains sensitive information.

Base Models vs Aggregated Models

Folder Name: FS_all and FS_agg, feature_selection

Then, the data goes through the base model of KNN, DT, GNB, MNB, MLP (1 hidden), MLP (2 hidden), SVM (linear), SVM (poly), SVM (rbf) and SVM (sigmoid). At this stage, all features are used. In the first table, other feature selection methods also take place, including Anova and Chi square, Anova and Information Gain, SFS and SBS. Only the normalised data table has higher accuracy, so starting forward, all data used is normalised. The data also goes through aggregation to reduce dimensionality. This occurs by obtaining the data's average (after using encoding to assign the results as numbers) based on the type of survey questions. There are 5 types: S (scale), P (perceived behavioural control of vaccine), A (benefit of the vaccine), B (perceived susceptibility of the person) and C (perceived barrier). After replacing the survey questions with the average, the data is used to go through the same ML models as above.

Normalized

Base Models:

	KNN	DT	GNB	MNB	MLP (1 hidden)	MLP (2 hidden)	SVM (linear)	SVM (poly)	SVM(rbf)	SVM (sigmoid)
Parameter	n_neighbors=10	max_depth=5, criterion=entropy	-	-	hidden_layer_size=(50,)	hidden_layer_size=(5, 80)	c=0.1	C: 1, degree: 2, gamma: scale	c=10, gamma=scale	c=1, gamma=scale
Booster - All Features	0.775	0.825	0.4	0.775	0.71	0.69	0.725	0.775	0.825	0.675
Booster - ANOVA + Chi2	0.825	0.775	0.9	0.825	0.775	0.65	0.72	0.78	0.78	0.68
Booster - ANOVA + IG	0.825	0.8	0.675	0.825	0.75	0.675	0.65	0.78	0.75	0.62
SFS	0.85	0.8	0.8	0.775	0.675	0.675	0.775	0.7	0.75	0.675
SBS	0.75	0.8	0.825	0.75	0.65	0.675	0.8	0.875	0.875	0.75

Aggregate Models:

Parameter	n_neighbors=10	max_depth=4, criterion=entropy	-	-	hidden_layer_size=(30,)	hidden_layer_size=(5, 40)	c=0.1	C:1, degree:3, gamma:'auto'	c= 10, gamma=scale	c=10, gamma=auto
Booster - v1 (B_AGG)	0.575	0.65	0.375	0.675	0.72	0.7	0.65	0.68	0.6	0.68
B_AGG - ANOVA + Chi2	0.675	0.65	0.825	0.725	0.7	0.775	0.7	0.8	0.75	0.72
B_AGG - ANOVA + IG	0.675	0.625	0.725	0.8	0.775	0.75	0.75	0.7	0.75	0.68
B_AGG - SFS	0.625	0.575	0.8	0.675	0.7	0.7	0.75	0.6	0.725	0.55
B_AGG - SBS	0.525	0.65	0.35	0.625	0.675	0.65	0.75	0.7	0.6	0.55

Unnormalized

	KNN	DT	GNB	MNB	MLP (1 hidden)	MLP (2 hidden)	SVM (linear)	SVM (poly)	SVM(rbf)	SVM (sigmoid)
Parameter	n_neighbors=4	max_depth=5, criterion=entropy	-	-	hidden_layer_size=(95,)	hidden_layer_size=(5, 85)	c=0.01	C: 10, degree: 3, gamma: scale	c=1, gamma=auto	c=1, gamma=auto
Booster - All Features	0.7	0.825	0.475	0.7	0.73	0.7	0.7	0.8	0.625	0.675
Booster - ANOVA + Chi2	0.6	0.775	0.9	0.8	0.575	0.675	0.7	0.6	0.65	0.68
Booster - ANOVA + IG	0.6	0.75	0.6	0.75	0.675	0.6	0.68	0.6	0.62	0.68
SFS	0.625	0.8	0.85	0.675	0.675	0.625	0.675	0.55	0.8	0.625
SBS	0.475	0.725	0.65	0.725	0.675	0.575	0.7	0.7	0.6	0.7
Booster - v1	0.425	0.65	0.4	0.525	0.375	0.6	0.7	0.57	0.62	0.62

Note: The subsequent steps use only the normalized data, which has the better results.

Ensemble Learning

Folder Name: ensemble-learning

The models with the best accuracy is used in the second table, which uses ensemble learning. The ensemble learning methods are random forests, bagging (which uses the best models as base estimator), boosting and XGBoost.

Base Models:

	RF	Bagging (DT)	Bagging (MLP)	Bagging (SVM poly)	Bagging (SVM rbf)	Boosting	XGBoost
All Features	0.73	0.7	0.7	0.8	0.7	0.71	0.75
Booster - ANOVA + Chi2	0.75	0.7	0.775	0.675	0.7	0.75	0.75

Aggregate Models:

	RF	Bagging (GNB)	Bagging (SVM poly)	Bagging (MLP 2 hidden)	Bagging (SVM sigmoid)	Boosting	XGBoost
All Features - AGG	0.675	0.425	0.675	0.575	0.575	0.73	0.65
B_AGG - ANOVA + Chi2	0.775	0.625	0.675	0.7	0.575	0.83	0.75

Choosing the Models for Explainability

Since the aim of this experiment is to choose the most important features that decide if a healthcare worker would agree to a subsequent booster, the best models were chosen based on i) accuracy and ii) the number of features chosen during feature selection (if available).

The chosen models for explainability are:

Model type	Chosen Model(s)	Accuracy	Notes
Base model - ALL Features	Decision tree	0.83	
	SVM (RBF)	0.83	
ALL Features with Anova + Chi Square	GNB	0.9	
Base model - Aggregated	Bagging (SVM poly)	0.8	This uses ensemble learning.
Aggregated Features with Anova + Chi Square	Boosting	0.83	This uses ensemble learning.
	GNB	0.83	

Note: Although the accuracy for SBS is high, the number of features chosen were lower compared to ALL Features with Anova + Chi Square, so it is not suitable for the aim of this experiment.

Local XAI Fusion

Folder Name: fusion xai —> phase_4

Since local explainability focuses on individual predictions, SHAP local and LIME are first conducted separately for the selected machine learning models to obtain the local feature importance for each test instance. However, similar to traditional XAI methods, SHAP and LIME may produce inconsistent feature importance rankings for the same instance.

To reduce the inconsistency between both explainability methods, LocalWeightFusion is introduced. The fusion combines the SHAP and LIME explanations into a single fused explanation using weighted scores. For LIME, the prediction probability is used as the weight, while the LIME score represents the feature importance. For SHAP, the positive SHAP values are used as the feature importance scores. The fused explanation is then calculated to produce a final local feature importance ranking.

The local fusion results are then evaluated by comparing Fusion vs SHAP, Fusion vs LIME and SHAP vs LIME using Kendall Tau, RBO and Recall@K. The aim is to determine whether the fused explanations produce stronger consistency compared to using SHAP or LIME individually.

Local results:

	Fusion vs SHAP	Fusion vs LIME	LIME vs SHAP
Kendall Tau	0.518280	0.874308	0.392587
RBO	0.787592	0.921233	0.710990
Recall@10	0.766667	1.000000	0.766667
Recall@20	1.000000	1.000000	1.000000
Recall@30	1.000000	1.000000	1.000000

Global XAI Fusion

Folder Name: fusion xai —> phase_5

For global explainability, SHAP is first applied to each trained machine learning model to obtain the global feature importance across the entire dataset. Since multiple models are used, the SHAP rankings from different models may produce inconsistent global feature importance results.

To reduce this inconsistency, GlobalWeightFusion is introduced. The SHAP values from each model are combined using weighted aggregation, where the weights are based on the accuracy of each model. Models with higher accuracy contribute more strongly to the final fused global feature importance ranking.

The global fusion is conducted separately for the base machine learning models, ensemble learning models and the combined base + ensemble models. Then, the fused global rankings are compared against the statistical baseline obtained from ANOVA and Chi-Square feature selection.

Kendall Tau, RBO and Recall@K are used to evaluate the agreement between the statistical baseline, individual SHAP rankings and GlobalWeightFusion rankings. The aim is to determine whether the fused global explanations produce stronger consistency compared to the traditional SHAP rankings from individual models.

Global results:

	ALL: DT	ALL: SVM (rbf)	ALL + AC: GNB	AGG: Bagging, SVM (poly)	AGG + AC: Boosting	AGG + AC: GNB	Stats vs Mean of ML Methods	Stats vs Fusion
Kendall Tau	0.12	0.12	0.68	-0.22	0.02	0.02	0.123333	0.41
RBO	0.38	0.37	0.74	0.02	0.06	0.06	0.271667	0.31
Recall@10	0.20	0.10	0.70	0.00	0.00	0.00	0.166667	0.10
Recall@20	0.30	0.45	0.85	0.00	0.10	0.10	0.300000	0.35
Recall@30	0.47	0.47	0.97	0.07	0.23	0.23	0.406667	0.63

Results

Folder Name: fusion xai —> results —> global, local

The most important features for vaccine hesitancy can be found in the data folders in results —> global and results —> local. They are ordered based on amount of feature importance.

The results are stable since the fusion means for both global and local are nearly always higher (~4/5 times) than the means of the ML models, for i. Kendall Tau, ii. RBO and iii. Recall@K. [Refer to the 2 tables in Local and Global XAI Fusion above.]